

Neural ODEs

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ODEs

Neural
Networks

Neural ODEs

Neural Differential Equations

A Different(ial) Approach to Machine Learning

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Overview

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A Brief Introduction to Ordinary Differential Equations

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What is a differential equation?

- An equation that relates a function to one or more of its derivatives.
- Simplest Example:

$$\frac{dy}{dx} = y$$

What is the goal of "solving" this equation?

- We are trying to find a function that satisfies this equation.
- Instead of finding a numerical value like in Algebra, we are trying to find a **function**.

Example of solving a DE

$$\frac{dy}{dx} = y \rightarrow dx \frac{dy}{dx} = y dx \rightarrow dy = y dx \rightarrow \frac{1}{y} dy = dx \quad (1)$$

$$\int \frac{1}{y} dy = \int dx \rightarrow \ln |y| = x + c \rightarrow |y| = e^{x+c} \rightarrow y = ce^x \quad (2)$$

A Brief Introduction to Neural Networks

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What is a Neural Network?

- Universal function approximator.
- Have a set of parameters that are learned over time.

How does it learn?

- The model learns by computing the error of its output and performing **backpropagation**.
- Propagating error back through each layer of the network.

How is this similar to differential equations?

- In both cases we are trying to find a function that satisfies a certain set of constraints, and so these ideas are inherently linked to one another.

Residual Neural Networks

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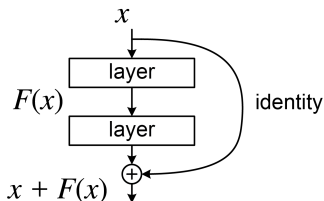
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What is a ResNet?

- It is a particular structure of neural network that processes sequential time data.



- It can process data at **discrete** time steps, and output data at a **discrete** time step.

Downsides?

- If we want a finer temporal resolution, the number of layers and complexity of the network increases.
- Is there a better solution?

Neural Ordinary Differential Equations

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Approach

- Approximate **derivative** rather than original function.
- We define the following ODE:

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, t, \theta) \quad (3)$$

- $\mathbf{x}$ is the function, and θ is an adjustable parameter.

Why is it useful?

- Allows for **continuous** time evaluation.
- Allows for unevenly spaced data without requiring more "layers".
- We can adjust the θ parameter in order for it to learn.

Learning

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How do we do backpropagation in normal ResNets?

- Calculate error and then take the gradient with respect to the previous layer output.
- To calculate parameter adjustment, we can use the Chain Rule:

$$\frac{dL}{d\theta} = \frac{dL}{d\mathbf{x}} \frac{d\mathbf{x}}{d\theta} \quad (4)$$

- Then, we repeat this process for each layer of the network, computing these values individually.

What about NODEs?

- How do we do this in a continuous format?

Continuous Backpropagation

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The Adjoint State

- We cannot go layer-by-layer
- Instead, we compute the **Adjoint State** by solving a second ODE backwards through time:

$$\frac{d\mathbf{a}}{dt} = -\mathbf{a}(t)^T \frac{\partial f(\mathbf{x}(t), t, \theta)}{\partial \mathbf{x}} \quad (5)$$

Where $\mathbf{a}(t) = \frac{\partial L}{\partial \mathbf{x}}$

The Parameter Gradient

- To calculate the error adjustment, we solve another similar ODE simultaneously.

$$\frac{\partial L}{\partial \theta} = \int_{t_1}^{t_0} -\mathbf{a}(t)^T \frac{\partial f(\mathbf{x}(t), t, \theta)}{\partial \theta} \quad (6)$$

Data

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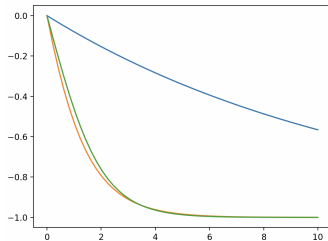
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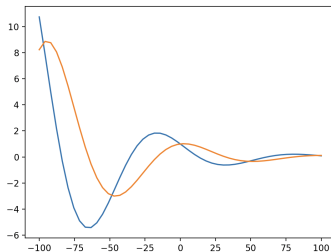
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Trained First Order NODE



Untrained Higher Order NODE



Possible Areas for Further Research

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Types of Differential Equations

- Research could be done into the applications and creation of neural differential equations using PDEs (Partial differential equations) and SDEs (Stochastic Differential Equations).

Different Function Structures

- The derivative is set equal to a function that can be modified or structured in different ways for different applications.

References

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Neural ordinary differential equations

<http://arxiv.org/pdf/1806.07366>

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The End